B4 Problems: Powers, roots and exponential growth

← B4 overview · Notes · Solutions

Unless a problem says otherwise: work without a calculator in problems 1–6 and 22–27; a calculator is allowed elsewhere. Give growth factors to at least 4 decimals and final answers rounded sensibly for the context.

Practice

1. Simplify (all letters are positive).

a) $a^5 \cdot a^3$ b) $\dfrac{x^7}{x^2}$ c) $(y^3)^4$ d) $(2p^3)^4$ e) $(3a^2b)^2 \cdot 2ab^3$

2. Calculate without a calculator.

a) 2^{-3} b) 5^0 c) $\left(\frac{2}{3}\right)^{-2}$ d) $(-2)^4$ and -2^4 e) $3^{-2} \cdot 3^5$ f) 10^{-4} as a decimal

3. Write as a single power of 2.

a) $8 \cdot 4^3$ b) $\frac{1}{16}$ c) $\sqrt{2}$ d) $32^{3/5}$ e) $4^{-3/2}$ f) $\sqrt[3]{16}$

4. Simplify.

a) $\sqrt{12}$ b) $\sqrt{50}$ c) $\sqrt{72}$ d) $\sqrt{18} + \sqrt{8}$ e) $\sqrt{3} \cdot \sqrt{12}$ f) $\sqrt[3]{54}$ g) $\sqrt{75} - \sqrt{27}$

5. Rationalise the denominator (*wortel uit de noemer halen*) and simplify.

a) $\dfrac{6}{\sqrt{3}}$ b) $\dfrac{1}{\sqrt{2}}$ c) $\dfrac{10}{\sqrt{5}}$ d) $\dfrac{\sqrt{2}}{\sqrt{6}}$ e) $\dfrac{2}{\sqrt{3}-1}$

6. a) Calculate: $27^{2/3}$, $16^{3/4}$, $32^{-2/5}$, $8^{-1/3}$, $\left(\frac{1}{9}\right)^{-1/2}$.

b) Write as a single power of x ($x > 0$): $\sqrt{x} \cdot x^2$, $\dfrac{1}{\sqrt[3]{x}}$, $\dfrac{x\sqrt{x}}{x^3}$.

7. Scientific notation.

a) Write $4\,500\,000$ and $0.000\,32$ in scientific notation.

b) Calculate $(3 \cdot 10^5) \cdot (4 \cdot 10^{-8})$ and $\dfrac{6 \cdot 10^{12}}{1.5 \cdot 10^4}$; give the answers in scientific notation.

c) Light travels $3.0 \cdot 10^8$ m per second. How many metres is a light-year (the distance light travels in 365.25 days)? Give your answer in scientific notation with 3 significant figures.

8. a) Give the growth factor for: an increase of 7%; a decrease of 12%; an increase of 150%.

b) Give the percentage change per period for the growth factors 1.035, 0.94 and 2.5.

9. For each table, decide whether it is linear, exponential or neither. Give a formula for the linear and exponential ones.

x	0	1	2	3	4
A	3	7	11	15	19
B	5	10	20	40	80
C	200	180	162	145.8	131.22
D	1	4	9	16	25

10. a) A population grows by 20% per day. Find the growth factor per week and per hour.

b) An investment grows by 5% per year. By what percentage does it grow in 10 years? And per month?

11. Consider the power functions $y = x^2$, $y = x^3$, $y = x^4$, $y = x^{-1}$, $y = x^{-2}$ and $y = \sqrt{x}$.

a) Which of their graphs are symmetric in the y-axis? Which are point-symmetric in the origin? Explain using $(-x)^n$.
b) Which graphs have asymptotes? Name them.
c) Which of the graphs pass through the point $(2, \frac{1}{4})$? Which through $(-1, -1)$?
d) Explain without a calculator why $x^4 < x^2$ for $0 < x < 1$.

12. a) 12 workers need 10 days to build a shed. How long do 15 workers need? And 8 workers? (Assume everyone works equally fast.)

b) Show that x and y in this table are inversely proportional (*omgekeerd evenredig*) and give a formula.

x	1	2	3	4	6
y	24	12	8	6	4

c) Name the asymptotes of the graph of your formula.

School-exam level

13. Bacteria. In a lab, the number of bacteria in a dish is given by $N = 500 \cdot 1.8^t$, with t in hours.

a) How many bacteria are there after 5 hours?
b) By what percentage does the number grow per hour?
c) Calculate the growth factor per minute (4 decimals).
d) After how many hours does the number first exceed 1 000 000? Answer to one decimal, using a table.
e) Find the doubling time in minutes, rounded to the nearest minute.

14. Medicine. A patient takes 400 mg of a medicine. The body breaks it down exponentially with a half-life (*halveringstijd*) of 6 hours.

a) How much is left after 24 hours?
b) Calculate the growth factor per hour, and the percentage by which the amount decreases per hour.
c) Give a formula for the amount A (mg) after t hours.
d) The medicine no longer works below 10 mg. After how many whole hours is the amount below 10 mg?

15. Savings. Sanne puts €2500 in a savings account with 3.2% interest per year (interest is added each year).

a) How much is in the account after 10 years?
b) After how many whole years is there at least €4000?
c) Sanne's brother says: "3.2% per year is 32% per 10 years." Explain why he is wrong and give the correct percentage.
d) Calculate the equivalent percentage per month.

16. Depreciation. A new car costs €32 000. Its value decreases by 15% per year.

a) What is the value after 4 years?
b) After how many whole years is the value below €10 000 for the first time?

c) By what percentage has the value decreased in total after 3 years?

17. Population. A town had 24 000 inhabitants in 2015 and 27 600 in 2020. Assume exponential growth.

a) Calculate the growth factor per 5 years and per year, and the percentage growth per year.
b) Give a formula for the population P, with t in years since 2015.
c) Predict the population in 2030.
d) How many inhabitants did the town have in 2010, if the same growth applied before 2015?

18. Measurements. A biologist counts the number of fruit flies in a jar every two days.

t (days)	0	2	4	6	8
N	120	173	249	358	516

a) Show that the growth is (approximately) exponential.
b) Find the growth factor per 2 days and per day.
c) Give a formula for N.
d) Estimate the number of fruit flies after 3 days.

19. Ferris wheel. A seat on a Ferris wheel (*reuzenrad*) is at its lowest point at $t = 0$. The table gives its height h (m) above the ground, with t in seconds.

t (s)	0	10	20	30	40	50	60	70	80
h (m)	2.0	7.9	22.0	36.1	42.0	36.1	22.0	7.9	2.0

The wheel keeps turning at the same speed.

a) Give the period, the equilibrium line (*evenwichtsstand*) and the amplitude.
b) What is the diameter of the wheel, and how high is the axle?
c) What is the height of the seat at $t = 250$?
d) Sketch the graph for $0 \le t \le 160$. Use it to estimate how many seconds per turn the seat is higher than 30 m.

20. Tides. In a harbour the water level is measured relative to NAP (*Normaal Amsterdams Peil*). The level is periodic with a period of 12 hours 25 minutes. On a certain day there is high water of 1.2 m above NAP at 03:00, and the next low water is 0.8 m below NAP.

a) Give the equilibrium line and the amplitude.
b) At what time is the first low water?
c) At what times are the next two high waters after 03:00?
d) At about what time between 03:00 and the first low water is the water level at the equilibrium line?

21. Light. The light intensity I (in lux) at distance d metres from a lamp is $I = \dfrac{400}{d^2}$.

a) Calculate I for $d = 2$ and $d = 4$.
b) What happens to I when the distance is tripled? Explain with the rules for powers.
c) At what distance is the intensity 25 lux?
d) What does the horizontal asymptote of the graph mean in this context?

Challenge

22. Which is bigger? No calculator.

a) 2^{300} or 3^{200}?

b) $\sqrt[10]{10}$ or $\sqrt[3]{2}$?
c) 2^{100} or 3^{63}? (Hint: $2^{10} = 1024$, $3^{21} = 10\,460\,353\,203$. Expanding $(1+x)^{10}$ as a product of ten brackets may help.)

23. Last digits.

a) What is the last digit of 7^{2027}?
b) What is the last digit of $2^{2027} + 3^{2027}$?

24. More irrational numbers.

a) Prove that $\sqrt{3}$ is irrational.
b) Where exactly does the same argument fail if you try it on $\sqrt{4}$?
c) Prove that $\sqrt{2} + \sqrt{3}$ is irrational. (You may first prove that $\sqrt{6}$ is irrational.)

25. Exponential equations. Solve exactly.

a) $4^x + 4^x + 4^x + 4^x = 2^{20}$
b) $2^x \cdot 4^{x+1} = 32^{x-2}$
c) $9^x = 3 \cdot 27^{x-1}$

26. Nested radicals. Write as $a + b\sqrt{c}$ or $\sqrt{m} + \sqrt{n}$, and check by squaring.

a) $\sqrt{3 + 2\sqrt{2}}$ b) $\sqrt{7 - 4\sqrt{3}}$ c) $\sqrt{11 + 6\sqrt{2}}$ d) $\sqrt{5 + 2\sqrt{6}}$

27. Big powers. Simplify $\dfrac{2^{2027} + 2^{2025}}{2^{2026} - 2^{2024}}$.

28. Testing the rule of 72. For an interest rate of $p\%$ per year, the rule of 72 says that the doubling time is about $72/p$ years.

a) For $p = 2, 6, 9, 24$, find (with a calculator and a table) the exact doubling time to two decimals, and compare with $72/p$.
b) For which of these rates is the rule best? For which worst?
c) Show that with 1% interest the doubling time is about 70 years (between 69 and 70). What does this suggest about using 70 instead of 72 for small rates?